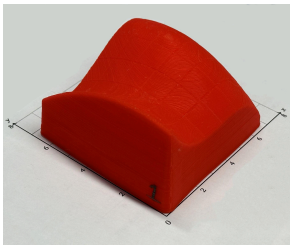
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3D Printed PLA Graph Model



A 3-dimensional model of a function's graph made out of polylactic acid (PLA) through 3D printers.

Materials and Costs:

- 3D Printer: ~$300 CAD. Many public libraries and universities have 3D printers you can use/rent for an hourly rate. We used MakerBot Sketch Large 3D printers.
- PLA filament: ~$25 CAD per roll.

Process of Prototyping:

- We used Stepan Paul's design, available on Thingiverse: https://www.thingiverse.com/thing:3072108
- We prototyped many different sizes of the model. Because ours were going to be used in a group setting, where 3-4 students were going to be sharing a model, we decided on a 70 mm side length.
- Bright filament colours (i.e. not black, grey, or white) make it easier to see details of the graph.
- We added a raft to the bottom of each model, which acted like a supportive base. This ensured that each model was uniform to each other, as the raft minimized 3D printer errors.
- We had infill at 10%, rather than the default 15%, which reduced printing time.
- We created a file that printed 4 models at once. Although each printing job took 4x as much time, it was useful for maximizing overnight printing sessions.

Step by Step:

1. Upload your desired STL file to an editing software connected to your printer. Here you can change the print size and settings. We used MakerBot CloudPrint web app: **Sketch Large.**
2. Upload your desired STL file to a flash drive, or a device that can be input to your 3d printer.
3. Insert the drive into the printer, and follow the specific directions of printing corresponding to your 3D printer. In our case, we had to first calibrate the printer with a piece of paper and then start the print job.
4. After the print job was finished we had to lightly remove the print from the 3D printer tray, and remove the raft we added from the model.

Tips and Tricks:

- Ask if your local library/university has a remote service to print models overnight if you're in a hurry to print.
- As the printing job starts, stay back for 15-30 min to check if the print has any errors. You can often mitigate or cancel the job if an error is found early on.

Photos:



